UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 008 · Problems on Taylor's and Maclaurin's series · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

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2. 1 · Taylor expansion about a point (a, b)
3. 2 · Maclaurin expansion about the origin
4. 3 · Approximate values from a Taylor polynomial
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A Formula sheet

1 Taylor expansion about a point (a, b)

E1.1 · Taylor's series about (a, b), to second order ESA ★★★

$$f(x,y) = f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \ldots, \quad h = x-a,\ k = y-b$$

GIVES	The polynomial to substitute into.
APPLIES	All derivatives evaluated at (a, b).
NOTE	h, k are shorthand added for clarity.

S008·3 · S008·14 · T008 00:00:37 · T008 00:01:25

E1.2 · Third-degree term

ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

GIVES The terms for "up to third degree".

APPLIES Add to E1.1 or E2.1.

NOTE Coefficients 1, 3, 3, 1; asked in four of six ESA questions.

S008·10 · ESA Dec-23 1b · ESA Jun-24 1c

2 Maclaurin expansion about the origin

E2.1 · Maclaurin's series (about the origin)

ESA ★★★

$$f(x,y) = f(0,0) + \{x f_x + y f_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}\} + \dots$$

GIVES The polynomial about $(0,0)$.

APPLIES Derivatives at $(0,0)$.

NOTE Slide 008·10 calls this "Taylor's series about (0,0)"; it is the same thing.

S008·10 · S008·12 · T008 00:06:58

E2.2 · One-combination check

ESA ★★

$$f = g(t),\ t = px + qy \implies f(x,y) = g(0) + g'(0)\,t + \frac{g''(0)}{2!}t^2 + \frac{g'''(0)}{3!}t^3 + \dots$$

GIVES The same expansion in one line.

APPLIES f depends on x and y only through $px + qy$.

NOTE $\log(2x + y + 1) = t - \frac{t^2}{2} + \frac{t^3}{3}$ with $t = 2x + y$; $e^{x-y} = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}$ with $t = x - y$.

ADDED FOR CLARITY S008·6 · S008·11

3 Approximate values from a Taylor polynomial

E3.1 · Approximation by a Taylor polynomial

ESA ★★

$$f(a+h, b+k) \approx f(a,b) + h f_x + k f_y + \frac{1}{2!}\{h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}\}$$

GIVES A numerical estimate.

APPLIES h, k small; (a,b) a point where f and its derivatives are easy.

NOTE Choose (a,b) with "nice" values: $(1,-1)$ for $(0.9,-1.2)$.

S008·7 · S008·9 · T008 00:12:10

B Symbol glossary

Symbol	Name	Meaning	Appears in
(a, b)	Expansion point	$(1, \frac{\pi}{4})$, $(1, \frac{\pi}{2})$, $(1, 1)$, $(1, -1)$ or $(0, 0)$ in class 008	E1.1
$h,\ k$	Increments	$h = x - a$, $k = y - b$ (shorthand added for clarity)	E1.1, E3.1
$f_x, \ldots, f_{yy}$	Partial derivatives	Evaluated at the expansion point	E1.1, E2.1
$f_{xxx}, \ldots, f_{yyy}$	Third partials	Coefficients 1, 3, 3, 1	E1.2
t	Combination	$2x + y$, $x - y$, $x + 2y$ in the one-variable check	E2.2
s	Step along a line	Figure G3.1's parameter only	G3.1
$a,\ b$ (in $e^{ax} \sin by$)	Constants	Not the expansion point here (it is the origin)	W2.5

Remark. $h = x - a$ and $k = y - b$ are shorthand added for clarity; the slides write $(x - 1)$, $(y - \frac{\pi}{4})$ and so on in full. "Up to three terms" (slide 008·5) means three non-zero terms, as the lecture explains; it is not the same as "up to third degree".